

**Started on** Friday, 14 August 2026, 4:07 PM**State** Finished**Completed on** Friday, 14 August 2026, 4:17 PM**Time taken** 9 mins 40 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
  {  
    for (i = 1; i <= num; ++i)  
    {  
      if (num % i == 0)  
      {  
        printf("%d ", i);  
      }  
    }  
  }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>  
2 int main(){  
3     int i,num;  
4     int count=0;  
5  
6     scanf("%d",&num);  
7  
8     for (i = 1; i <= num; ++i)  
9     {  
10        count++;  
11        if (num % i == 0)  
12        {  
13            count++;  
14        }  
15        count++;  
16    }count++;  
17    printf("%d",count);  
18 }
```

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            printf("%d ", i);
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Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

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Answer:

```
1 #include<stdio.h>
2 int main(){
3     int i,num;
4     int count=0;
5
6     scanf("%d",&num);
7
8     for (i = 1; i <= num;++i)
9     {
10         count++;
11         if (num % i== 0)
12         {
13             count++;
14         }
15         count++;
16     }count++;
17     printf("%d",count);
18 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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